Arsalene Khachmadi

Data Scientist

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Profile

Newly graduated from EURECOM and ESPRIT, specialised in data science, with hands-on experience building data pipelines for fraud detection at Amadeus (PySpark, Scala, Databricks). Strong skills in ML modelling, NLP, and LLMs, with a growing interest in applying AI to healthcare and life sciences. Comfortable in agile, multicultural environments and committed to responsible AI development.

Experience

Amadeus **03/2025 – 08/2025**

Data Engineer intern *Nice, France*

- Analysed business requirements across 2 fraud detection teams and translated operational needs into ML-ready problem definitions and data models.
- Performed feature engineering to extract and construct meaningful predictive signals from raw transactional data, improving model input quality and granularity.
- Trained, and fine-tuned supervised ML models for fraud detection, and using evaluation metrics (precision, recall, F1)
- Collaborated in agile sprints with business and technical teams.
- Stack: Python, PySpark, Scala, Databricks, Elasticsearch, Confluence, Git, Agile/Scrum.

3S Globelnet **06/2023 – 08/2023**

Data Engineer intern *Ariana, Tunisia*

- Collected and structured heterogeneous data via web scraping and integration into Microsoft Azure.
- Built ETL pipelines (Python) for product data preparation, reducing manual processing time significantly.
- Benchmarked 3+ supervised and unsupervised ML models to inform data-driven product decisions.

Education

EURECOM **09/2023 – 09/2025**

Master in Intelligent Communication Systems (Data & IA) *Sophia-Antipolis, France*

ESPRIT **09/2021 – 09/2023**

Engineering Degree in computer science – Data Science specialisation *Ariana, Tunisia*

Université Centrale Privée **09/2018 – 06/2021**

Bachelor Degree in computer science *Tunis, Tunisia*

Academic projects

Comparative Study of LLMs and PLMs | *Python, NLP, Transformers, GPT, BERT, T5* **March–June 2024**

- Extracted and analysed causal relations in socio-political event corpora using NLP pipelines.
- Applied prompt engineering and data augmentation to improve model performance on noisy corpora.
- Quantitatively benchmarked 4 models (GPT, BERT, Zephyr-7B, T5) on F1 and precision metrics.

Skills-Based Curriculum Platform | *Python, NLP, classification de texte, PowerBI* **Feb.–May 2023**

- Scraped LinkedIn to extract skills from 500+ professional profiles.
- Applied a text classification task on the raw data we collected and built a platform for automatic curriculum updates.

Hard skills

Langages: Python, SQL, Scala, Java, R, MATLAB, C#
MLOps & AI: Scikit-learn, Deep Learning, NLP, LLMs (OpenAI GPT, BERT, Zephyr-7B, T5), TensorFlow
Data Engineering & Cloud: PySpark, Databricks, Spark, Hadoop, Elasticsearch, SQL Server, Azure Data Lake, GCP
Tools: Docker, CI/CD (GitHub Actions), Git, PowerBI, Jupyter, Flask

Languages

Arabic: Native **French:** Fluent **English:** Fluent **German:** Basic